

Archytas of Tarentum: Doubling the Cube and Linking Music to Mathematics

Archytas of Tarentum was an influential mathematician in ancient Greece, renowned for both his political leadership and mathematical achievements [1, p. 3]. He was born c. 428 BCE in Tarentum, now Taranto, Italy and died c. 350 BCE of unknown causes [1, p. 5]. Unlike many mathematicians of his time, Archytas was deeply involved in politics and was reported to have been elected general of Tarentum seven times. He was apparently very successful in his leadership and helped the city prosper [2, VIII.79]. If true, this shows how admired he was for his time since the office could not usually be held more than once [2, VIII.79]. Archytas also had many connections to other mathematicians, being closely associated with Plato and even in some accounts saved Plato when he fell into political trouble during a visit to Syracuse, helping arrange his release [3]. Archytas was also credited with mechanical inventions, supposedly creating a wooden pigeon that could fly using compressed air, considered to be one of the earliest recorded mechanical devices if true [4, § 10.12].

Archytas was a second-generation follower of Pythagoras and often considered one of the most important of the later Pythagoreans [1]. Being Pythagorean involved emphasising that numbers and mathematical relationships underlie the structure of the universe. According to this view, numerical proportion governs nature and can be observed in many natural phenomena [5]. One of the most famous examples of this comes from music; the Pythagoreans discovered that harmonious musical intervals correspond to simple numerical ratios between the lengths of vibrating strings [6, pp. 50-51]. This suggested that harmony could be explained mathematically, reinforcing the idea that it forms the underlying principles of nature. For the Pythagoreans, mathematics was not only a discipline but a philosophical framework for understanding the world since studying ratios, proportions and numerical relationships was thought to reveal the hidden structure of the universe [5]. Archytas' work reflects the Pythagorean tradition [1], particularly in his work on ratios and geometry, and often combining the two to provide mechanical proofs revolutionary for his time.

The Delian problem, also known as the doubling or duplication of the cube, is one of the three classical geometric problems of ancient Greek mathematics along with angle trisection and squaring the circle [7] [8, pp. 134-140]. The problem consists of having a cube and wanting to construct another with exactly twice the volume, which was especially difficult to ancient Greek mathematicians due to their affinity for constructions only using a compass and straightedge formalised in Euclid's Elements. It is now known that it is impossible to construct with just these tools from Pierre Wantzel's proof in 1837 [9]. Despite this, Archytas is credited as the first to solve this problem by Eutocius, said to have done it with an ingenious geometric construction [10, p. 245]. In modern notation the problem is straightforward. If a cube has side length a , its volume is $V = a^3$. To construct a cube with twice the volume we need a side length x such that $x^3 = 2V = 2a^3$ giving $x = a\sqrt[3]{2}$. Archytas and the other ancient Greek mathematicians did not have this algebraic notation and therefore had to get creative and use geometry.

Archytas first used a result from Hippocrates, who showed that doubling the cube can be reduced to finding two mean proportions between two lengths, a mean proportional for a and b being an x such that $a : x = x : b$, if the original cube has side a , take segment $b=2a$ and find x and y such that $a : x = x : y = y : 2a$, [10, p. 200]. We can verify this using modern algebra to get $\frac{a}{x} = \frac{x}{y} = \frac{y}{2a}$ and from the first pair of ratios get $x^2 = ay$ and from the second get $y^2 = 2ax$, so now from the first rearrange to get $y = \frac{x^2}{a}$, then substitute into the second to get $(\frac{x^2}{a})^2 = 2ax$, Then $\frac{x^4}{a^2} = 2ax$, $x^3 = 2a^3$, Archytas' goal was to find the segment x such that this ratio holds and doubles the cube. Archytas used an

incredible geometric construction to find them by constructing three geometric surfaces whose intersection determines the required mean proportional. The surfaces being two quadratic surfaces (a cylinder and cone) and a more complex curved surface generated by rotating a circle, now known as a torus-like surface. Finding the intersection on all the surfaces satisfies the constraints of the ratios and Archytas wanted to find this point [11].

Archytas started with three points, O, A and B in a plane such that $OB = 2OA$, with a circle drawn with diameter OB and OA being the original side length of the cube we want to double [11]. Then, bringing this into three dimensions, Archytas considered a family of circles perpendicular to the plane OAB with diameter OB highlighted in red in Figure 1 that fanned out by rotation, generating a curved torus-like surface. Archytas then constructed a cone by drawing straight lines from O through points on these circles labeled Q_θ in Figure 2, this is the point N_θ . Then a cylinder was constructed by extending the original circle D, meaning all three surfaces are constructed and the intersection point on the surfaces determines the segment x that satisfies the mean conditions, meaning the cube with side x has twice the volume of the cube with side OA. The mathematical tools used by Archytas for this construction are the Pythagorean theorem, properties of right triangles and intersecting chords theorem that appear in Elements [12].

Figure 3 shows a rough 3D render of the construction made by myself and gives a general visualisation of what is happening in Figures 1 and 2. This construction survives history through the commentary of Eutocius [13] and is what historians consider to be one of the earliest known mechanical solution in 3D geometry [10], even having Plato supposedly criticize him for bringing mechanical ideas into geometry, saying that him and Eudoxus were “corrupters and destroyers of the pure excellence of geometry” [14, pg. 473], demonstrating how revolutionary and controversial his method was at the time. Interestingly, a similar relationship appears in modern music theory by the major third in equal temperament, a musical interval that is exactly one third of an octave that multiplies the frequency of a tone by $2^{4/12} = 2^{1/3} = \sqrt[3]{2}$, the side length of the doubled cube [15].

Archytas frequently related mathematics with music theory, reflecting the Pythagorean view that musical harmony could be explained through numerical ratios [5]. Ptolemy even identifying that Archytas “engaged in the study of music most of all the Pythagoreans” [16]. One important type of ratio Archytas studied was superparticular ratios, which have the form $\frac{n+1}{n}$, important since they describe many musical intervals and times [16]. Archytas proved these ratios cannot be divided into a mean proportional. In modern notation, if $a:b = (n+1):n$, there exists no number x such that $a:x = x:b$. Assuming such an x exists gives $\frac{a}{x} = \frac{x}{b}$, then $x^2 = ab$ but if $a:b = (n+1):n$, this implies $x^2 = n(n+1)$. Because consecutive integers are coprime, their product cannot be a perfect square, giving

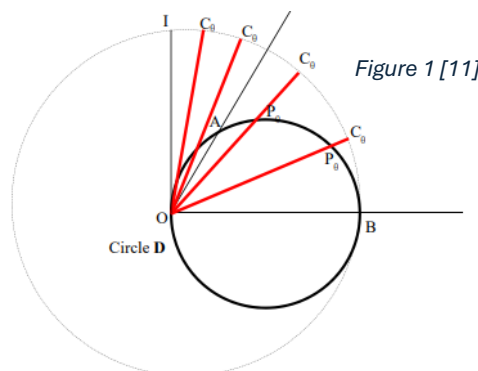


Figure 1 [11]

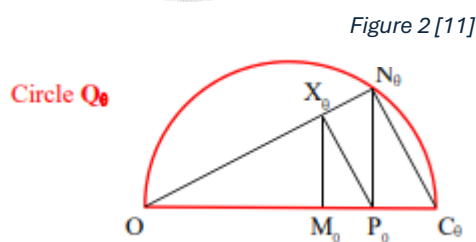


Figure 2 [11]

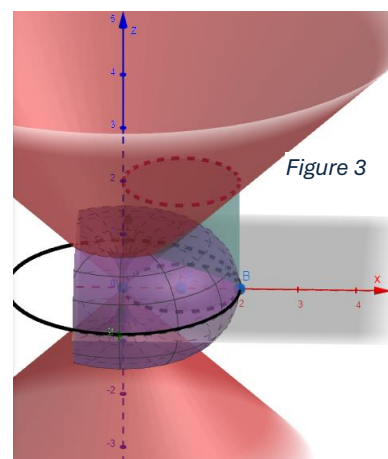


Figure 3

a contradiction [10, p. 90]. Later commentators on his work noted that this proof assumes properties of numbers that were not rigorously justified in the argument itself [10]. This proof was still important in Greek music theory, where many important musical scales correspond to a superparticular ratio such as 2:1 (the octave), 3:2 (fifth), 4:3 (fourth) and 9:8 (the whole tone), as well as times, most famously $\frac{5}{4}$, $\frac{4}{3}$, $\frac{3}{2}$ [16]. Archytas' proof was later tweaked by Euclid in *Elements*, Book VII to be considered a full ancient Greek rigorous proof [12].

Archytas stands out among other early Greek mathematicians due to his forward and innovative thinking. His solution to the Delian problem demonstrated his originality and breadth of thinking, being one of the earliest known three-dimensional mechanical solutions in Greek geometry. His work on superparticular ratios shows his Pythagorean identity and the connection between music and mathematics. His work was an essential step in the development of Greek mathematics, showing how abstract concepts for the time can be used to understand geometric problems and the structure of music.

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